PC2174A: Mathematical Methods in Physics I
Assignment 2

1. [10 pts]

(a) If $y(x) = \exp(-x^2)$, show that for $n \geq 1$,

$$y^{(n+1)}(x) + 2x\, y^{(n)}(x) + 2n\, y^{(n-1)}(x) = 0.$$

(b) The Bessel function of order 0, $y \equiv J_0(x)$, satisfies the second-order ordinary differential equation

$$x^2 \frac{\mathrm{d}^2 y(x)}{\mathrm{d}x^2} + x \frac{\mathrm{d}y(x)}{\mathrm{d}x} + \left(x^2 - n^2\right) y(x) = 0$$

for $n = 0$.

(i) Given that $J_0(0) = 1$, find $J_0'(0)$.

(ii) Find $J_0''(0)$.

Answer:

(a) Notice that

$$y' = -2xe^{-x^2} = -2xy \;\Rightarrow\; y' + 2xy = 0 \tag{1}$$

Differentiate the equation above n times,

$$y^{(n+1)} + (2xy)^{(n)} = 0 \tag{2}$$

where

$$(2xy)^{(n)} = \sum_{i=0}^{n} \binom{n}{i} \frac{\mathrm{d}^i}{\mathrm{d}x^i}(2x) \frac{\mathrm{d}^{(n-i)}}{\mathrm{d}x^{(n-i)}} y = 2xy^{(n)} + 2ny^{(n-1)}$$

As the result,

$$y^{(n+1)} + 2xy^{(n)} + 2ny^{(n-1)} = 0 \tag{3}$$

(b) Suppose the Bessel function can be expressed as a serise:

$$J_0(x) = \sum_{n=0}^{\infty} a_n x^n, \quad a_0 = 1 \tag{4}$$

According to the differential equation, we have:

$$\sum_{i=0}^{\infty}(i+1)(i+2)a_{i+2}x^{i+2} + \sum_{i=0}^{\infty}(i+1)a_{i+1}x^{i+1} + \sum_{i=0}^{\infty} a_i x^{i+2} = 0 \tag{5}$$

Rearranging the above expression, we get

$$a_1 x + \sum_{i=0}^{\infty}[(i+1)(i+2)a_{i+2} + a_i]x^{i+2} = 0 \tag{6}$$

Since x is arbitrary, $a_1 = 0$, which implies:

$$J_0'(0) = a_1 = 0 \tag{7}$$

And considering the term of x^2, also since x is arbitrary, the coefficient should be 0

$$2a_2 + a_0 = 0 \;\Rightarrow\; a_2 = -\frac{1}{2} \tag{8}$$

which implies

$$J_0''(0) = -\frac{1}{2} \tag{9}$$

2. [10 pts]
The frequency of vibrations of a vibrating violin string is given by

$$f = \frac{1}{2L}\sqrt{\frac{T}{\mu}},$$

where L is the length of the string, T is the tension and μ is its average linear mass density.

(a) Find the rate of change of frequency with respect to

(i) the length L (when T and μ are constant),

(ii) the tension T (when L and μ are constant), and

(iii) the linear mass density μ (when L and T are constant).

(b) The pitch of a note (how high or low the note sounds) is determined by the frequency. The higher the frequency, the higher the pitch. Determine what happens to the pitch of a note

(i) when the effective length of a string is decreased by placing a finger on the string so a shorter portion of the string vibrates,

(ii) when the tension is increased by turning a tuning peg,

(iii) when the linear mass density is increased by switching to another string.

Answer:

(a) Just differentiate the equation provided

$$\text{(i)}\frac{\mathrm{d}f}{\mathrm{d}L} = -\frac{1}{2L^2}\sqrt{\frac{T}{\mu}};\ \text{(ii)}\frac{\mathrm{d}f}{\mathrm{d}T} = \frac{1}{4L\sqrt{T\mu}};\ \text{(iii)}\frac{\mathrm{d}f}{\mathrm{d}\mu} = -\frac{1}{4L}\sqrt{\frac{T}{\mu^3}} \tag{10}$$

(b)

(i) When the effective length of a string is decreased, the frequency is increased, and the pitch is higher;

(ii) When the tension is increased, the frequency is increased, and the pitch is higher;

(iii) When the linear mass density is increased, the frequency is decreased, and the pitch is lower.

3. [10 pts]
Throughout this problem, we consider particle dynamics in one dimension. In Newtonian mechanics, the linear momentum p of a particle with mass m is defined by $p \equiv mv$, where v is the velocity of the particle. Newton's second law of motion is

$$F = \frac{\mathrm{d}p}{\mathrm{d}t},$$

where F is the total force acting on the particle. If the mass remains constant, this becomes $F = ma$, where $a = \frac{\mathrm{d}v}{\mathrm{d}t}$ is the acceleration.
In Einstein's special relativity, the relativistic momentum of a particle of mass m is

$$p = \frac{mv}{\sqrt{1 - v^2/c^2}},$$

where v is the velocity of the particle and c is the speed of light in vacuum. Experiments show that $F = \frac{\mathrm{d}p}{\mathrm{d}t}$ is still valid in relativistic mechanics provided that the relativistic momentum is used.

Answer:

$$F = \frac{\mathrm{d}p}{\mathrm{d}t} = \frac{m\sqrt{1 - v^2/c^2}\,\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{mv^2/c^2}{\sqrt{1-v^2/c^2}}\frac{\mathrm{d}v}{\mathrm{d}t}}{1 - v^2/c^2} = \frac{m\frac{\mathrm{d}v}{\mathrm{d}t}}{\left(1 - \frac{v^2}{c^2}\right)^{3/2}} \tag{11}$$

4. [20 pts]

A curve in the plane can be described as the graph of a function or an equation. A more convenient way is to think of it as the path of a moving particle whose position is changing over time. Then, each of the x- and y-coordinates of the particle's position becomes a continuous function of a third variable t, called a parameter:

$$x = f(t), \qquad y = g(t).$$

The set of points $(x, y) = (f(t), g(t))$ defined by these equations is a parametric curve.

(a) A curve in the plane is parameterized as

$$x = a\cos 3\theta, \qquad y = a\cos\theta,$$

where a is a constant.

(i) Obtain an expression for $\frac{\mathrm{d}y}{\mathrm{d}x}$ on the curve.

(ii) Eliminate the parameter θ to find a Cartesian equation of the curve. Then obtain the expression for $\frac{\mathrm{d}y}{\mathrm{d}x}$ from the Cartesian equation.

(iii) Verify that the expressions for $\frac{\mathrm{d}y}{\mathrm{d}x}$ obtained above are equivalent.

(iv) Show that the origin is a point of inflection. Is it a critical point?

(v) Use the information gained above to sketch the curve, paying particular attention to its shape near the points $(-a, a/2)$ and $(a, -a/2)$ and to its slope at the 'end points' (a, a) and $(-a, -a)$.

(b) The parametric equations for the motion of a charged particle released from rest in electric and magnetic fields at right angles to each other take the form

$$x(\theta) = a(\theta - \sin\theta), \qquad y(\theta) = a(1 - \cos\theta),$$

where a is a constant.

(i) Find the tangent to the curve.

(ii) Use the result above at a few calculated values of x and y to sketch the form of the particle's trajectory.

Answer:

(a)

(i)

$$\mathrm{d}x = -3a\sin 3\theta \mathrm{d}\theta,\ \mathrm{d}y = -a\sin\theta \mathrm{d}\theta \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\sin\theta}{3\sin 3\theta} \tag{12}$$

(ii)

$$\cos 3\theta = 4\cos^3\theta - 4\cos\theta \tag{13}$$

As a result

$$x = 4a\cos^3\theta - 3a\cos\theta = \frac{4}{a^2}y^3 - 3y \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \left(\frac{\mathrm{d}x}{\mathrm{d}y}\right)^{-1} = \frac{a^2}{12y^2 - 3a^2} \tag{14}$$

(iii)

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{a^2}{12a^2\cos^2\theta - 3a^2} = \frac{1}{9 - 12\sin^2\theta} \tag{15}$$

And

$$\frac{\sin\theta}{3\sin 3\theta} = \frac{\sin\theta}{6\sin\theta(1 - \sin^2\theta) + 3(1 - 2\sin^2\theta)\sin\theta} = \frac{1}{9 - 12\sin^2\theta} \tag{16}$$

(iv)

$$\left.\frac{\mathrm{d}^2x}{\mathrm{d}y^2}\right|_{x=0} = \left.\frac{24}{a^2}y\right|_{y=0} = 0,\ \left.\frac{\mathrm{d}x}{\mathrm{d}y}\right|_{x=0} = \left.\left(\frac{12}{a^2}y^2 - 3\right)\right|_{y=0} = -3 \tag{17}$$

As a result, the origin is a point of inflection and not a critical point.

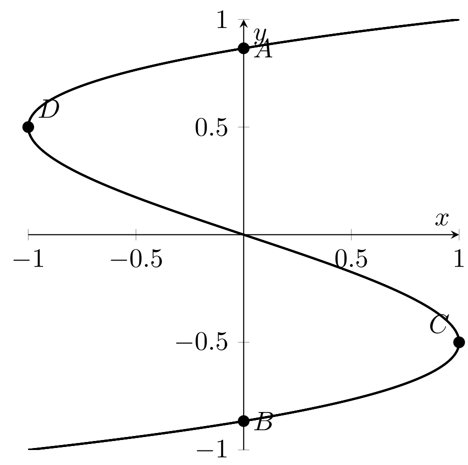

Figure 1: $x = a\cos 3\theta$, $y = a\cos\theta$ with key points., a=1

(v) First, the zeros of the curve (zeros here means $x = 0$) are $A(0, \frac{a\sqrt{3}}{2})$, $B(0, -\frac{a\sqrt{3}}{2})$; Second, the critical points ($\frac{\mathrm{d}x}{\mathrm{d}y} = 0$) are $C(a, -a/2)$, $D(-a, a/2)$. The curve is drawn in Figure 1:

(b) (i)

$$\mathrm{d}x = (a - a\cos\theta)\ \mathrm{d}\theta,\ \mathrm{d}y = \sin\theta\ \mathrm{d}\theta \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \cot\frac{\theta}{2} \tag{18}$$

For a given $x_0,\ \theta_0,\ y_0$, the tangent to the curve is

$$l : x = \tan\frac{\theta_0}{2}(y - y_0) \tag{19}$$

(ii)

$$x(\theta + 2\pi) = a(\theta + 2\pi - \sin(\theta + 2\pi)) = x(\theta) + 2a\pi \tag{20}$$

So we can just consider $x \in [0, 2a\pi]$; zeros are $A(0,0),\ B(2a\pi, 0)$, and the critical point is $C(a\pi, 2a)$,and the curve can be drawn:

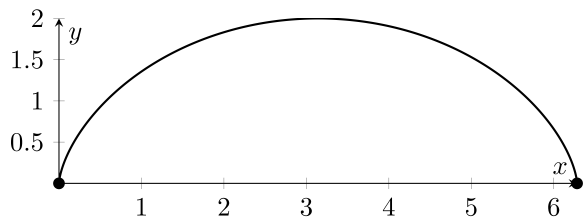

Figure 2: $x(\theta) = a(\theta - \sin\theta)$, $y(\theta) = a(1 - \cos\theta)$ with key points A, B, C., a=1

5. [15 pts]

Neglecting air resistance, the path of a projectile, moving not too far above the ground under gravity, is a curve in a two-dimensional plane. Denoting the plane as the xz-plane where the x-axis is horizontally rightward and the z-axis is vertically upward, the path of the projectile is given by

$$\begin{cases} x(t) = x_0 + u_0 t, \\ v_x(t) = u_0, \\ a_x(t) = 0, \end{cases} \qquad \begin{cases} z(t) = z_0 + v_0 t - \frac{1}{2}gt^2, \\ v_z(t) = v_0 - gt, \\ a_z(t) = -g, \end{cases}$$

where u_0 and v_0 are the initial velocities along the x- and z-directions respectively, x_0 and z_0 are the initial positions, $g \approx 9.80\,\mathrm{m/s^2}$ is the magnitude of the acceleration due to gravity, and t is the time.

(a) Derive an equation for the projectile's path in terms of x and z. Show that the path is a parabola.

(b) Calculate the angle of flight $\theta(t)$ and the rate of loss of kinetic energy $K(t)$ at time t.

(c) A projectile is fired up an incline (incline angle ϕ) with an initial speed v_i at an angle θ_i with respect to the horizontal ($\theta_i > \phi$) as shown in the figure.

(i) Find the distance d, measured along the incline, from the launch point to the point where the projectile strikes the incline.

(ii) For what value of θ_i is d a maximum and what is that maximum value?

Answer:

(a)

$$t = \frac{1}{u_0}(x-x_0),\ z = z_0+\frac{v_0}{u_0}(x-x_0)-\frac{1}{2}\frac{g}{u_0^2}(x-x_0)^2 = -\frac{g}{2u_0^2}x^2+\left(\frac{x_0 g}{u_0^2}+\frac{v_0}{u_0}\right)x+z_0-\frac{v_0 x_0}{u_0}-\frac{g x_0^2}{2u_0^2} \quad (21)$$

Obviously, it is a parabola.

(b)

$$\tan\theta = \frac{v_z}{v_x} = \frac{v_0 - gt}{u_0} \Rightarrow \theta = \arctan\left(\frac{v_0}{u_0} - \frac{g}{u_0}t\right) \quad (22)$$

and the rate of loss of K is

$$\left|\frac{\mathrm{d}K}{\mathrm{d}t}\right| = \left|mv\frac{\mathrm{d}v}{\mathrm{d}t}\right| = \left|m\sqrt{u_0^2+(v_0-gt)^2}\cdot\frac{-(v_0-gt)g}{\sqrt{u_0^2+(v_0-gt)^2}}\right| = mg(v_0-gt) \quad (23)$$

(c)
(i) When the particle is hit on the incline, its coordinates have the relation:

$$\tan\phi = \frac{z}{x} = \frac{v_0}{u_0} - \frac{g}{2u_0}t \Rightarrow t = \frac{2v_0}{g} - \frac{2u_0}{g}\tan\phi \quad (24)$$

and the x is

$$x = \frac{2v_0u_0}{g} - \frac{2u_0^2}{g}\tan\phi \Rightarrow d = \frac{x}{\cos\phi} = \frac{2v_0u_0}{g\cos\phi} - \frac{2u_0^2}{g}\frac{\tan\phi}{\cos\phi} \quad (25)$$

where $v_0 = v_i\sin\theta_i$ and $u_0 = v_i\cos\theta_i$. As a result, given θ_i and ϕ, d is

$$d = \frac{v_i^2\sin 2\theta_i}{g\cos\phi} - \frac{2v_i^2\cos^2\theta_i\tan\phi}{g\cos\phi} \quad (26)$$

(ii) Differentiate equation (17) and suppose it equals to 0

$$\frac{\mathrm{d}d}{\mathrm{d}\theta_i} = \frac{2v_i^2\cos 2\theta_i}{g\cos\phi} + \frac{2v_i^2\sin 2\theta_i}{g}\frac{\tan\phi}{\cos\phi} = 0 \Rightarrow \tan 2\theta_i = -\cot\phi \quad (27)$$

As a result, in case:

$$\theta = \frac{\phi}{2} + \frac{\pi}{4} \quad (28)$$

d is a maximum and it is

$$d = \frac{v_i^2\sin\left(\phi+\frac{\pi}{2}\right)}{g\cos\phi} - \frac{\left(1+2\cos\left(\phi+\frac{\pi}{2}\right)\right)v_i^2\tan\phi}{g\cos\phi} = \frac{v_i^2}{g}\frac{1-\sin\phi}{\cos^2\phi} = \frac{v_i^2}{g(1+\sin\phi)} \quad (29)$$

6. [15 pts]
The curvature of a curve at a given point P is a measure of how quickly the curve changes direction at that point. The *circle of curvature* at a point P on a plane curve is the circle in the plane of the curve that

- is tangent to the curve at P;
- has the same curvature the curve has at P;

- has centre that lies toward the concave side of the curve.

The radius of the circle of curvature is called the *radius of curvature*. For a plane curve with equation $y = f(x)$, the radius of curvature is

$$\rho(x) = \frac{\left\{1 + [f'(x)]^2\right\}^{3/2}}{|f''(x)|}.$$

(a) Find the radius of curvature at the point (x, y) on the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Check the special case $b = a$ for which the ellipse becomes a circle.

(b) The curve whose equation is

$$x^{2/3} + y^{2/3} = a^{2/3},$$

where a is a constant, for positive x and y and which is completed by its symmetric reflections in both axes, is known as an *astroid*. Find its radius of curvature in the first quadrant.

Answer:

(a) For $y \geq 0$ (the same as $y \leq 0$), $y = \sqrt{b^2 - \frac{b^2}{a^2}x^2}$,

$$y' = \frac{-\frac{b^2}{a^2}x}{\sqrt{b^2 - \frac{b^2}{a^2}x^2}} \tag{30}$$

$$y'' = -\frac{\frac{b^2}{a^2}\sqrt{b^2 - \frac{b^2}{a^2}x^2} + \frac{\frac{b^2}{a^2}x^2}{\sqrt{b^2 - \frac{b^2}{a^2}x^2}} \cdot \frac{b^2}{a^2}x^2}{b^2 - \frac{b^2}{a^2}x^2} = \frac{\frac{b^4}{a^2}}{\left(b^2 - \frac{b^2}{a^2}x^2\right)^{3/2}} \tag{31}$$

the radius of curvature is

$$\rho = \frac{\left(1 + \frac{\frac{b^4}{a^4}x^2}{b^2 - \frac{b^2}{a^2}x^2}\right)^{3/2}}{\frac{b^4}{a^2}} \cdot \left(b^2 - \frac{b^2}{a^2}x^2\right)^{3/2} = \frac{a^2}{b^4}\left(b^2 + x^2\left(\frac{b^4}{a^4} - \frac{b^2}{a^2}\right)\right)^{3/2} \tag{32}$$

when a=b,

$$\rho = \frac{1}{a^2} \cdot a^3 = a \tag{33}$$

it is correct since the ellipse now turns to a circle.

(b)

$$\frac{2}{3}x^{-\frac{1}{3}}\mathrm{d}x + \frac{2}{3}y^{-\frac{1}{3}}\mathrm{d}y = 0 \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{y^{\frac{1}{3}}}{x^{\frac{1}{3}}} \tag{34}$$

and

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{y^{-\frac{1}{3}} + x^{-\frac{2}{3}}y^{\frac{1}{3}}}{3x^{\frac{2}{3}}} = \frac{a^{2/3}}{3x^{4/3}y^{1/3}} \tag{35}$$

The radius of curvature is

$$\rho = \frac{(1 + \frac{y^{2/3}}{x^{2/3}})^{3/2}}{a^{2/3}} \cdot 3x^{4/3}y^{1/3} = \frac{\frac{a}{x}}{a^{2/3}} \cdot 3x^{4/3}y^{1/3} = 3(axy)^{1/3} \tag{36}$$

7. [20 pts]
A rainbow is produced by the reflection of sunlight by spherical drops of water in the air. The primary rainbow is the type commonly seen in regions where rainbows appear. It is produced by light reflecting once inside the drops. The left figure below shows a light ray entering and then leaving a spherical drop after one internal reflection. The final direction of travel is deviated (turned) from the initial direction of travel by angular deviation Δ.

(a) Show that the angular deviation Δ, in radians, is given by

$$\Delta = 2\theta_i - 4\theta_r + \pi.$$

(b) The index of refraction n of an optical medium is defined by the ratio of the speed of light in vacuum to that in the medium, $n \equiv c/v$, where v is the speed of light in the medium and c is the speed of light in vacuum. Use Snell's law to express Δ in terms of θ_i and the index of refraction of the water n_w in the drop.

(c) A rainbow will form when the angular deviation Δ is *stationary* in the incident angle θ_i. If this condition is satisfied, all the rays with incident angles close to this critical angle will be sent back in the same direction, producing the bright zone in the sky. Find an equation which the critical angle must obey.

(d) The index of refraction in water is 1.342 for violet light and 1.330 for red light. Find the critical angle and angular deviation for violet and red light respectively. When you view the primary rainbow, which colour, red or violet, is higher above the horizon? Check that your results agree with the known results available on the internet.

(e) A *secondary rainbow* is formed when the incident light undergoes two internal reflections in a spherical drop of water as shown in the right figure above. Repeat the above calculations for the secondary rainbow. When you view a secondary rainbow, is red or violet higher above the horizon? Again, check that your results agree with the known results available on the internet.

Answer:

(a) The deviation from the first refraction is

$$\Delta_1 = \theta_i - \theta_r \tag{37}$$

from the first reflection is

$$\Delta_2 = \pi - 2\theta_r \tag{38}$$

from the second refraction is

$$\Delta_3 = \theta_i - \theta_r \tag{39}$$

So the total angular deviation is

$$\Delta = \theta_i - \theta_r + \pi - 2\theta_r + \theta_i - \theta_r = 2\theta_i - 4\theta_r + \pi \tag{40}$$

(b)Snell Laws shows:

$$n_w \sin\theta_r = \sin\theta_i \Rightarrow \theta_r = \arcsin\left(\frac{1}{n_w}\sin\theta_r\right) \tag{41}$$

As a result, the angular deviation is

$$\Delta = 2\theta_i - 4\arcsin\left(\frac{1}{n_w}\sin\theta_r\right) + \pi \tag{42}$$

(c) If the angular deviation is stationary in the incident angle:

$$\mathrm{d}\Delta = 2\mathrm{d}\theta_i - 4\mathrm{d}\theta_r \Rightarrow \frac{\mathrm{d}\theta_i}{\mathrm{d}\theta_r} = 2 \tag{43}$$

And from Snell Law, we have

$$n_w \cos\theta_r \mathrm{d}\theta_r = \cos\theta_i \mathrm{d}\theta_i \Rightarrow \frac{\mathrm{d}\theta_i}{\mathrm{d}\theta_r} = \frac{n_w\cos\theta_r}{\cos\theta_i} \tag{44}$$

We get the equation :

$$\frac{n_w\cos\theta_r}{\cos\theta_i} = 2 \Rightarrow 2\cos\theta_i = \sqrt{n_w^2 - \sin^2\theta_i} \tag{45}$$

As a result, critical angle is to be obeyed:

$$\theta_i = \arcsin \frac{4 - n_w^2}{3} \tag{46}$$

(d)

$$\theta_{violet} = \arcsin \frac{4 - 1.342^2}{3} \approx 47.1°, \quad \theta_{red} = \arcsin \frac{4 - 1.33^2}{3} \approx 48° \tag{47}$$

Red is over violet light, which the critical angles also imply ($48° > 47.1°$)

(e) The angular deviation for secondary rainbow is

$$\Delta^{(2)} = \theta_i - \theta_r + 2 \cdot (\pi - 2\theta_r) + \theta_i - \theta_r = 2\theta_i - 6\theta_r + 2\pi \tag{48}$$

Repeat the precedure we have done for the first rainbow, we have

$$\frac{n_w \cos\theta_r}{\cos\theta_i} = 3 \Rightarrow \theta_i = \arcsin \frac{9 - n_w^2}{8} \tag{49}$$

and the critical angles for violet and red light are

$$\theta_{violet} = \arcsin \frac{9 - 1.342^2}{8} \approx 64.14°, \quad \theta_{red} = \arcsin \frac{9 - 1.33^2}{8} \approx 64.83° \tag{50}$$

The critical angle of red light is still bigger than that of violet light. But the direction of critical angles in the second rainbow is opposite to the first rainbow, for secondary rainbow violet light is about the red light.

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